

Отдельные задачи с SEEMOUS

1. (2007) Given $a \in (0, 1) \cap \mathbb{Q}$ let $a = 0.a_1a_2a_3 \dots$ be its decimal representation. Define

$$f_a(x) = \sum_{n=1}^{\infty} a_n x^n, \quad x \in (0, 1).$$

Prove that f_a is a rational function of the form $f_a(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials with integer coefficients.

Conversely, if $a_k \in \{0, 1, 2, \dots, 9\}$ for all $k \in \mathbb{N}$, and $f_a(x) = \sum_{n=1}^{\infty} a_n x^n$ for $x \in (0, 1)$ is a rational function of the form $f_a(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials with integer coefficients, prove that the number $a = 0.a_1a_2a_3 \dots$ is rational.

2. Let F be a field and let $P : F \times F \rightarrow F$ be a function such that for every $x_0 \in F$ the function $P(x_0, y)$ is a polynomial in y and for every $y_0 \in F$ the function $P(x, y_0)$ is a polynomial in x .

Is it true that P is necessarily a polynomial in x and y , when a) $F = \mathbb{Q}$, the field of rational numbers? b) F is a finite field?

Prove your claims.

3. (2008) Let $P_0, P_1, P_2, \dots$ be a sequence of convex polygons such that, for each $k \geq 0$, the vertices of P_{k+1} are the midpoints of all sides of P_k . Prove that there exists a unique point lying inside all these polygons.
4. (2010) Inside a square consider circles such that the sum of their circumferences is twice the perimeter of the square.

a) Find the minimum number of circles having this property.

b) Prove that there exist infinitely many lines which intersect at least 3 of these circles.

5. (2011) Given vectors $\bar{a}, \bar{b}, \bar{c} \in \mathbb{R}^n$, show that

$$(\|\bar{a}\| \langle \bar{b}, \bar{c} \rangle)^2 + (\|\bar{b}\| \langle \bar{a}, \bar{c} \rangle)^2 \leq \|\bar{a}\| \|\bar{b}\| (\|\bar{a}\| \|\bar{b}\| + |\langle \bar{a}, \bar{b} \rangle|) \|\bar{c}\|^2,$$

where $\langle \bar{x}, \bar{y} \rangle$ denotes the scalar (inner) product of the vectors $\bar{x}$ and $\bar{y}$ and $\|\bar{x}\|^2 = \langle \bar{x}, \bar{x} \rangle$.

6. (2015) For any positive integer n , let the functions $f_n : [0, 1] \rightarrow [0, 1]$ be defined by $f_n(x) = f_1(f_{n-1}(x))$, where $f_1(x) = 3x - 4x^3$. Solve the equation $f_n(x) = 0$.
7. (2026) Find all bijective functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$x^n f(x) + y^n f(y) \geq 2x^n f(y)$$

for all $x, y \in \mathbb{R}$, where n is a fixed positive integer.